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# Literature Review Instructions

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## 1 Abstract

Concise summary of purpose, methods, key results, and contribution.

Highlight uniqueness: “linking TESM with real-estate listings.”

## 2 Introduction and Motivation

The Tasmanian Enterprise Suitability Map (TESM) is a geospatial decision-support tool that integrates soil, climate, and landscape data to classify land as well suited, suitable, marginally suitable, or unsuitable for a range of agricultural enterprises. Developed by the Tasmanian Institute of Agriculture (TIA) in collaboration with industry, TESM supports planning and investment by providing science-based assessments of crop potential across Tasmania. By making suitability information publicly accessible through LISTmap, it has become a cornerstone of agricultural planning and resource allocation in the state.

Motivation: current LISTMap/real-estate platforms are passive & fragmented.

Despite its strengths, existing platforms present two key limitations. First, they respond poorly to changes in agribusiness trends, including the emergence of new crops that may become economically viable or strategically important for food security in the near future. Second, they are not directly linked with the property market, leaving a critical gap between geospatial crop suitability analysis and the practical realities of land acquisition. Prospective buyers or agribusiness investors seeking to identify land parcels that align with their crop preferences must manually consult multiple sources, including property listing sites, agronomy reports, and local council data. This process is time-consuming, fragmented, and often discourages comprehensive due diligence.

Prospective buyers and landholders in Tasmania therefore lack an efficient mechanism to assess which crops are suitable for newly listed properties at the time they appear on the market. Current real-estate platforms do not integrate geospatial factors such as soils, climate, topography, or water access into actionable summaries, meaning that opportunities are frequently missed. This project addresses that gap by automatically assessing crop suitability for new property sales listings and directly linking users with properties that match their preferred crops. To achieve this, the project develops an integrated system that combines property listing information, cadastral parcel boundaries, and statewide crop suitability layers. This system computes parcel-level suitability and presents the results through an interactive Shiny dashboard, providing users with a more personalised and efficient way to evaluate investment opportunities.

Building on this problem statement, the present study redefines the role of TESM by moving beyond the static functions of LISTmap. The project develops an integrated system that links live property sales data with crop suitability assessments derived from TESM, providing users with a more personalised and efficient way to identify land that matches their preferred crops. By combining property listing information, cadastral parcel boundaries, and statewide crop suitability layers, the system computes parcel-level suitability and delivers a more dynamic and actionable decision-support tool.

The significance of this research extends across several domains. For investors, the system enables better decision-making by reducing business risk and enhancing opportunity discovery. It assists those who already have specific crop strategies by identifying parcels that match their requirements as soon as they are listed, while also supporting landholders or capital investors who wish to explore which crops are most suitable for a given parcel. For the **agribusiness sector**, it provides a scalable framework for integrating suitability analysis into real-time land acquisition decisions, enabling faster and more confident responses in an environment of rising land values, shifting crop demands, and climate uncertainty. By reducing the risk of costly misallocation, the system ensures that capital and resources are directed toward opportunities with the greatest long-term viability. For **policy and planning bodies**, the system demonstrates how public geospatial resources such as TESM can be repurposed into practical, user-focused applications that bridge information gaps between government datasets and private-sector decision-making. It also highlights that these resources are not confined to a small niche of agricultural specialists, but can deliver value to mainstream business communities, including property-based investors and developers. This broader relevance strengthens the case for continued improvement of TESM and LISTmap through the inclusion of additional variables and more sophisticated modelling approaches. By connecting suitability assessment with property market dynamics, this project contributes to the creation of a more responsive, transparent, and data-driven agricultural investment environment in Tasmania.

### 3 Objective

This project aims to integrate TESM crop suitability layers with live property listing data in order to generate parcel-level suitability assessments. Building on this foundation, it develops an interactive Shiny dashboard that allows users to view ranked property recommendations, examine limiting factors, and directly access the corresponding property listings. The project also evaluates the tool's functionality through case studies in Tasmania and explores its potential for broader application, extending to regions with comparable suitability datasets and to future industry-facing implementations.

## 4 Literature Review

### 4.1 The Tasmanian Enterprise Suitability Map (TESM)

The [Tasmanian Enterprise Suitability Map \(TESM\)](#) represents a mature, data-driven framework for assessing agricultural potential across Tasmania. Originating from the [Wealth from Water](#) program initiated in 2010 by the Tasmanian Department of Primary Industries, Parks, Water and Environment, which has since been restructured and renamed as the Department of Natural Resources and Environment Tasmania ([NRE Tas](#)) in collaboration with the Tasmanian Institute of Agriculture (TIA) and the University of Sydney, TESM sought to provide an objective, reproducible, and spatially explicit basis for agricultural development planning. The program was itself a direct response to the Innovation Strategy (2009) led by Jonathan West, which identified irrigation expansion as a transformative opportunity for Tasmania's regional economy. The pilot phase initially covered about 70,000 hectares in the Meander Valley and Midlands, subsequently expanding to state-wide coverage through initiatives such as [Water for Profit](#). In this progression, the emphasis shifted from qualitative soil surveys to Digital Soil Mapping (DSM) and Digital Soil Assessment (DSA) techniques that operationalised soil and climate data into decision-support tools for land managers and investors (Kidd, 2020).

TESM's analytical foundation rests upon the integration of three primary data domains, consisting of soil, climate, and topography. The domains are processed through quantitative spatial modelling. The soil component draws on more than 900 core samples describing attributes such as pH, clay content, drainage class, and stone content. [Climate inputs](#) originate from a network of over 270 temperature sensors distributed across representative terrain positions, enabling fine-scale modelling of variables such as frost risk, chill hours, growing-degree days, and extreme heat frequency. Topographic variables, derived from Shuttle Radar Topography Mission (SRTM) data, gamma-radiometric imagery, and surface geology, capture slope, aspect, and elevation effects. These heterogeneous datasets are harmonised into raster layers at 30 to 80 m resolution and modelled using regression-tree and regression-kriging approaches to produce spatially continuous soil-attribute predictions with validation accuracies appropriate for regional-scale land evaluation (Kidd & Webb, 2015). The result is a three-dimensional digital representation of the Tasmanian soil landscape that supports downstream suitability analyses.

The suitability evaluation itself employs a deterministic, rule-based framework derived from the Food and Agriculture Organization (FAO, 1976) land-evaluation system and refined through the “most-limiting-factor” principle articulated by Klingebiel and Montgomery. Each crop or enterprise is governed by a rule set developed jointly by TIA researchers and industry experts, defining optimal and limiting parameter thresholds across the soil–climate–topography continuum. The model classifies

each grid cell into four main hierarchical categories, i.e. *well suited* , *suitable* , *marginally suitable* , *unsuitable* based on the lowest-rated criterion. Thus, if drainage or frost risk is poor even when all other factors are optimal, the overall classification defaults to “unsuitable.” This conservative logic enhances interpretability and ensures that biophysical limitations are not overlooked in investment decisions. The framework also differentiates between *manageable* and *unmanageable* constraints, acknowledging that interventions such as liming or artificial drainage can upgrade a site’s effective suitability (Kidd et al., 2015).

Following early operational success, the project advanced to produce state-wide 3D soil-attribute maps at 80 m resolution, later refined to 30 m, which were integrated into the national [Terrestrial Ecosystem Research Network \(TERN\)](#) and [GlobalSoilMap.net](#) repositories (Kidd & Webb, 2015). Complementary climatic projections from [Climate Futures for Tasmania](#) were incorporated to generate suitability layers for 2030 and 2050 under [Representative Concentration Pathway \(RCP\) 8.5 scenarios](#). Each enterprise is therefore represented by three temporal surfaces of current (2018), near-future (2030), and long-term (2050), enabling preliminary adaptation planning. An aggregated [Enterprise Versatility Index \(EVI\)](#) summarises the number of viable enterprises/crops per grid cell, thereby identifying land parcels capable of diversified production. The integration of these high-resolution soil and climate grids has substantially improved Tasmania’s ability to visualise spatial heterogeneity in agricultural potential.

Despite its technical sophistication, several important limitations are recognised in both scholarly evaluations and official documentation. First, spatial reliability varies with data density: accuracy is highest in intensively surveyed agricultural districts and lowest in sparsely sampled regions such as Tasmania’s southwest (Kidd & Webb, 2015). Second, the model presumes the universal availability of irrigation infrastructure, intended to decouple soil–climate potential from temporary water scarcity, yet one that may lead to over-optimistic classifications in non-irrigated areas (Kidd, 2020). Third, its deterministic thresholding system does not propagate measurement or model uncertainty; a marginal deviation in a single input variable can therefore cause a disproportionate downgrade in the final suitability score. This contrasts with probabilistic or Bayesian approaches that attach confidence intervals to each class boundary. Fourth, while the inclusion of multiple climate scenarios partially addresses temporal change, TESM remains a “snapshot” model rather than a dynamic yield-forecasting tool that reflects inter-annual variability or adaptive management. Furthermore, the consistency of scoring among crop enterprises has raised concerns regarding inter-scale reliability. Some rule sets employ integer scales (1.0 – 4.0), whereas others use fractional increments (1.0, 1.1, 2.0 etc.), complicating cross-crop comparison and aggregation.

**Table 1:** *List of All Current Crops in TESM*

| Type                   | Crops                                                                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cereals                | barley, linseed, wheat                                                                                                                                              |
| perennial horticulture | blueberriesNHB, blueberriesSHB, cherries, hazelnuts, olives, raspberries, sparklingwg, strawberries, tablewg                                                        |
| vegetables             | carrots, carrotseed, onions, potatoes                                                                                                                               |
| pharmaceuticals        | hemp, poppies, pyrethrum                                                                                                                                            |
| pastures               | cocksfootcontinental, cocksfootmediterranean, lucerne, phalaris, redclover, ryegrass, strawberryclover, tallfescuecontinental, tallfescuemediterranean, whiteclover |
| forestry               | E_globulus_tree, E_nitens_tree, P_radiata_tree                                                                                                                      |

Another limitation is the restricted flexibility for new or specialty crops. TESM currently covers around 33 enterprises, but adding further crops requires formal updates by custodians at the Department of Natural Resources and Environment Tasmania. The process of adding new crops also remains administratively constrained, as formal approval by custodians at the Department of Natural Resources and Environment Tasmania (NRE Tas) is required. These constraints collectively limit the model's responsiveness to emerging commodities or evolving market priorities. Nevertheless, TESM's transparent logic and documented metadata provide strong interpretability, allowing users to understand how each suitability class is determined and which variables exert the greatest influence.

The Tasmanian wine sector provides a salient example of TESM's applied relevance. Building on earlier climate modelling (Webb et al., 2015, 2018), the Enterprise suitability mapping and Tasmania's wine sector (Brullo et.al, 2024) demonstrates how TESM outputs guide site selection for viticulture by combining frost-risk surfaces, growing-degree-day thresholds, and soil-drainage indices. These analyses have supported Tasmania's transition toward high-value sparkling and cool-climate table wines, where irrigation expansion and land-use zoning are strategically coordinated with biophysical suitability. TESM data are increasingly used by policy agencies to outline priority irrigation districts and agricultural investment zones, reinforcing its dual role as a scientific and economic planning instrument.

In summary, the Tasmanian Enterprise Suitability Map embodies one of the most comprehensive, operational applications of digital soil assessment in Australia. Its integration of DSM, deterministic rule-based evaluation, and climate projection modelling enables evidence-based agricultural planning and investment. However, the framework's reliance on fixed thresholds, irrigation assumptions, and

limited adaptability underscores the need for more flexible, uncertainty-aware, and user-interactive systems. Overall, TESM remains a robust first-pass planning tool that transforms complex soil–climate datasets into interpretable land-suitability products while simultaneously revealing opportunities for future methodological refinement and broader integration with market-based decision systems. These design choices position TESM within the broader spectrum of suitability approaches, where alternative frameworks introduce probabilistic treatment of thresholds and machine-learning hybrids.

## 4.2 Approaches to Crop Suitability Mapping

Beyond deterministic rulesets, newer approaches either quantify uncertainty explicitly or learn interactions from data. Crop-suitability modelling spans a spectrum from deterministic, rule-based classifications to probabilistic and hybrid frameworks. Deterministic systems remain attractive because their thresholds and limiting factors are transparent and easy to audit, but they risk brittleness when small measurement errors or local idiosyncrasies push a site across a class boundary. Malone and Kidd outline how conventional land-evaluation pipelines can be extended to quantify and propagate uncertainty using probabilistic thresholds, fuzzy membership functions, and Monte-Carlo style perturbations, so that maps communicate confidence as well as class labels. They also emphasise that “most-limiting-factor” logic, while interpretable, should be complemented by methods that express the likelihood of class membership, particularly where input layers carry nontrivial prediction error. (Malone & Kidd, 2015)

**Table 2:** *List of All Variables in TESM*

| Class          | Variables                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| climate        | Air temperature, Chill hours, Daily maximum temperature, Extreme heat risk, Frost risk, Growing Degree Days, Growing Season Temperature, Heat stress, Hot weather during summer, Rainfall |
| soil attribute | Depth to sodic layer, Duplex soil, Electrical conductivity, Exchangeable calcium, Exchangeable magnesium, Soil depth, Soil Depth, Soil drainage, Soil texture, Stone abundance            |
| topography     | Elevation, Slope                                                                                                                                                                          |

Against this backdrop, the Tasmanian DSA/ESM tradition exemplifies a deterministic ruleset applied to gridded soil and climate covariates, with an explicit “most-limiting-factor” aggregation to derive the overall class per pixel. Rules are compiled from agronomic literature and expert workshops, thresholds are evaluated per attribute, and the lowest (i.e., most limiting) attribute determines the final class. This architecture delivers high interpretability and aligns well with stakeholder expectations about

“what limits what,” but it also motivates complementary representations (e.g., fuzzy limits) when classes hinge on narrow cut-offs or when inputs have known error bands. (Kidd et al., 2015)

A parallel stream of work explores machine-learning and hybrid methods that retain expert structure while learning nonlinear interactions from data. Open-source frameworks such as [CropSuite v1.0](#) combine modular crop response functions, climate/soil drivers, and flexible scoring to simulate suitability under alternative scenarios. The platform shows how open architectures can encode expert knowledge yet remain extensible as new covariates or crops are added, an attractive property when emerging commodities or climate adaptations are under consideration. (Zabel, 2022; Taghizadeh-Mehrjardi et al., 2020; Abdelrahman, 2021) National-level example further illustrate design choices around resolution, transparency, and uncertainty. The [New Zealand S-Map](#) that linked suitability work for perennial crops integrates phenology-aware climate indicators (e.g., bud-break timing) with frost-risk functions. This feature avoids single-indicator brittleness by combining risk components (frost and bud-break) so that delayed phenology does not falsely appear “low-risk” in very cold regions. Scenario analyses then trace suitability shifts under RCP pathways, making explicit how scores move as climate drivers change—a template for communicating both present-day classification and future risk envelopes (Vetharaniam et al., 2020; SLMACC Project 405422, 2020).

At the global scale, the FAO Global Agro-Ecological Zones (GAEZ) framework ([Agro-MAPS](#)) evaluates suitability, potential yields, and yield gaps by integrating harmonised soils/terrain, climate (historical and scenario-based), land cover and crop calendars, and agricultural statistics within a multi-module system (agro-climate, biomass/yield, edaphic constraints, integration/downscaling). Uncertainty is handled via scenario modelling (e.g., RCP pathways, CO<sub>2</sub> fertilisation) and by providing ranges for yields and constraints. Compared with TESM’s state-level deterministic rules (and S-Map’s national, phenology-aware risk composition discussed above), GAEZ offers much broader spatial scope and scenario-rich foresight, but at coarser resolution and with less direct farm-scale applicability. (FAO GAEZ v4 Documentation, 2022)

Taken together, contemporary suitability mapping is converging on hybrids: rule-based logic for transparency; probabilistic or fuzzy treatment for uncertainty; and modular, open tools for extensibility. Case studies from Tasmania (deterministic, interpretable rules over high-resolution inputs), New Zealand (phenology-aware, scenario-tested climate risk composition), and national zonation work (policy-ready aggregation) collectively suggest a practical path: retain expert-defined constraints where they are well established, but surface uncertainty and scenario sensitivity wherever inputs, thresholds, or management assumptions are weak. This orientation both improves scientific robustness and yields outputs that are directly actionable for investors, planners, and end-users who must make decisions under uncertainty (Malone & Kidd, 2015; Vetharaniam et al., 2020; Benson, 2016; Zabel,



2022). Yet even the most robust models must be delivered through interfaces that non-specialists can interrogate; this is the role of modern GIS-enabled decision support.

### 4.3 Geographic Information Systems (GIS) and Spatial Decision Support

Web-based GIS and [SDSS](#) operationalise those models, turning layered rasters and rules into auditable, interactive decisions. Geographic Information Systems (GIS) form the analytical backbone of modern spatial decision-making, enabling the integration of diverse environmental, climatic, and socioeconomic data into unified analytical frameworks. Carver (2015) articulated this transition from early stand-alone GIS models to participatory, web-enabled spatial decision support systems (SDSS), emphasizing how web-based GIS/multi-criteria evaluation (MCE) systems allow wider public access and enhance transparency in spatial planning. The shift from desktop-based GIS to web-delivered applications addressed a central limitation of earlier decision tools by allowing distributed users and policy stakeholders to interact directly with spatial layers and evaluation criteria online. This transformation established the conceptual foundation for modern geospatial dashboards and real-time mapping interfaces used in environmental management and land-use allocation.

Recent technological developments have further advanced GIS toward integrated, real-time territorial intelligence systems. The Trasimeno Smart Land initiative in central Italy exemplifies this evolution, extending Smart City principles into rural landscapes by combining 3D geospatial data, IoT sensors, and citizen-reported information through an integrated territorial platform (PITER) (Bianconi et al, 2025). The project operationalizes the Digital Twin concept for regional governance, integrating 24 thematic layers to monitor infrastructure, public works, and land-use change dynamically. Its modular and interoperable architecture ensures high data quality and scalability, providing a replicable model for rural or fragmented territories. This case underscores how GIS, when paired with continuous sensor networks and participatory reporting, can transform static mapping into a living decision-support environment for sustainable land management.

Parallel progress is evident in sector-specific web-based tools. Kim et al. (2025) developed the Web-based Rainfall Erosivity Calculation (WREC) tool, which integrates minute-interval rainfall datasets with simplified estimation models to generate high-resolution erosivity maps across South Korea. By embedding a rainfall-erosivity algorithm into a web interface, WREC enables planners and agricultural engineers to derive accurate soil-erosion risk indicators without specialized programming expertise. This demonstrates the power of web-based GIS systems to translate complex environmental models into accessible interfaces, fostering data democratization and reproducible environmental planning.

Similarly, Deval et al. (2022) introduced [Pi-VAT \(Prioritization-Visualization and Analysis Tool\)](#), a Shiny-based geospatial dashboard for hydrologic and watershed management. Built on the R

Shiny framework, Pi-VAT synthesizes output from process-based models such as WEPP and SWAT into interactive maps and plots, enabling managers to compare “what-if” scenarios across multiple watersheds and management treatments. The system’s use of open-source components, such as Leaflet, Plotly, and DataTables to illustrates how R’s geospatial ecosystem supports dynamic spatial analysis without proprietary software. Importantly, Pi-VAT bridges scientific modelling and stakeholder engagement, a critical step toward collaborative environmental governance.

The ExActR platform (Gibbons et al., 2025) represents a further refinement of this web-based paradigm, offering an open-source Shiny application for compiling ecosystem extent accounts under the UN System of Environmental Economic Accounting (SEEA-EA). ExActR enables users to load shapefiles, generate land-cover change tables, and visualize them through Leaflet maps and barplots—transforming complex geospatial workflows into intuitive dashboards. The integration of R packages such as *sf*, *terra*, and *shinydashboard* confirms that open-source GIS environments can support reproducible, scalable, and user-friendly decision tools aligned with international accounting standards.

Together, these studies illustrate a coherent trajectory: from static, expert-oriented GIS systems toward interactive, web-enabled, and participatory decision platforms. While accessibility and visualization have dramatically improved, challenges remain in ensuring interoperability among heterogeneous data sources and maintaining performance when handling high-resolution datasets. Nevertheless, the convergence of web-based GIS, real-time data integration, and open-source toolkits marks a decisive shift in environmental decision support. Within this evolving context, the current project’s Shiny-based dashboard builds directly on these advancements, extending the logic of Carver’s early web-SDSS vision and the functionality of tools such as Pi-VAT and ExActR to integrate Tasmania’s crop-suitability maps (TESM) with real-estate listings. In doing so, it transforms GIS from a back-end analytical engine into a front-end decision interface that connects agronomic suitability, spatial intelligence, and market accessibility for sustainable agricultural investment. However, most SDSS remain adjacent to the property market rather than embedded within it.

#### **4.4 PropTech and the Digitalisation of Land Information**

Property technology (PropTech) fills the market interface, including transactions, valuation, discovery, but rarely integrates agronomic evidence. The emergence of PropTech represents a significant digital transformation of global real-estate systems, driven by advances in automation, blockchain, and data analytics. The World Bank (2021) situates this trend within a broader evolution of land institutions toward transparency, accessibility, and decentralised data sharing, particularly in emerging markets where administrative inefficiencies and opaque valuation systems impede land investment. Similarly,

the International Federation of Surveyors (FIG, 2022) underscores that modern property systems increasingly depend on digital platforms and spatial interoperability, enabling more responsive and equitable land governance. These developments collectively redefine how property data are recorded, exchanged, and verified—marking a shift from static, document-based registries toward dynamic digital infrastructures that underpin both valuation and development planning.

While PropTech systems have achieved notable efficiency gains in transaction management, valuation, and urban property marketing, their applicability to agricultural or biophysical land assessment remains limited. A recent bibliometric and systematic review of digital real estate shows that research and tool development cluster around information-communication technologies, smart-city applications, data networking, and decision systems for urban markets: emphasising amenities, transactions, and platformisation, rather than integrating agronomic indicators such as soil productivity, topography, or irrigation access (Naeem et al., 2023).

The acceleration of PropTech adoption was notably amplified during the COVID-19 pandemic. Lee and Lee (2022) empirically demonstrate that online real-estate activity in Singapore surged as lockdowns restricted physical inspections, compelling both agents and buyers to rely on digitised market signals such as virtual tours and automated valuation models. Their study further observes a strong co-movement between online listing data and market price dynamics, underscoring that PropTech has become an essential medium for real-time market participation and risk management. Yet, such dependence on digital listings also intensifies exposure to user-generated biases, as information reliability varies according to the platform's verification protocols.

In this context, Meister and Reinholtz (2025) reveal a crucial behavioural limitation of online rating systems. Through analysing over 1.5 million Airbnb reviews, they found that platform-endorsed quality certifications (e.g., “Superhost”) paradoxically reduced subsequent user ratings, as consumers adjusted expectations upward and consequently rated experiences more harshly. This phenomenon demonstrates that user-generated content is inherently contextual and psychologically biased, challenging the notion that aggregated digital feedback provides an objective reflection of quality. Applied to property listings, it implies that relying solely on user reviews or seller-provided information could distort perceived land value or suitability, especially where biophysical quality is not independently verified through scientific data layers.

To counter such subjectivity, recent PropTech research explores blockchain-based valuation systems as a mechanism for enhancing transparency and trust. Adilime et al. (2024) developed “PropChain,” a prototype blockchain application built on Hyperledger Fabric, demonstrating that decentralised ledgers can reduce client influence and data manipulation in property valuation processes. By

distributing immutable valuation records across multiple stakeholders, blockchain enforces traceable accountability and prevents unilateral alteration of property data. This innovation aligns with earlier theoretical propositions by Levy and Schuck (2005), which argued that information transparency is the most effective safeguard against valuation bias. Although the Nigerian pilot study identified challenges in data verification and institutional readiness, it confirmed blockchain's potential as a neutral intermediary—a principle equally transferable to agricultural land transactions where transparency of soil, climate, and zoning data is critical.

In summary, contemporary PropTech innovations—from AI-driven market analytics to blockchain verification—collectively redefine the landscape of property information systems. Yet, their integration with biophysical and agronomic datasets remains underdeveloped. The reviewed literature indicates a persistent thematic gap: existing PropTech ecosystems primarily quantify market behaviour, while land-suitability frameworks quantify environmental potential. Bridging these two domains through interoperable, transparent, and user-centric digital infrastructures represents the next frontier of sustainable land governance. Within this context, the present project's Shiny-based decision-support platform directly addresses this integration gap—linking verified GIS-derived suitability data with live property-market interfaces to deliver an empirically grounded, trustworthy, and actionable PropTech model for agricultural investment. This asymmetry is visible in current Australian listing sites.

### **4.5 Real-Estate Listing Platforms and Agricultural Land Information**

An audit of mainstream property platforms in Australia (e.g., Domain, RealEstate.com.au) and rural-focused sites (e.g., Farmbuy, Farmproperty) shows indication to emphasise transactional attributes—parcel area and boundaries, buildings, tenure, access to services, and local amenity. These interfaces rarely expose verified biophysical layers (soil chemistry, drainage class, frost/chill metrics, slope), nor do they provide reproducible suitability logic. Where “suitable crops” are listed, claims are typically seller-supplied and lack traceable links to datasets or rule sets, leaving buyers to manually cross-reference external sources (e.g., government soil maps, climate rasters, council overlays). This workflow is fragmented and time-consuming for agribusiness users, and it systematically disadvantages non-GIS specialists who cannot efficiently assemble soil–climate–topography evidence for decision-making.

The absence of verifiable biophysical evidence on listings is not a trivial omission; it goes to the heart of what “suitability” means. Snelder et al. demonstrate that land-use suitability is not an intrinsic property of a parcel. Rather, suitability depends on landscape-level choices, environmental objectives, and the subjective weighting of indicators—hence any unverified, listing-level claim of “suitable for crop X” is, by construction, contestable without explicit data and methodological transparency. Their

framework shows that suitability emerges from composite indicators (productive potential, relative contribution to downstream effects, environmental pressure), each contingent on regional targets and modelled processes; as a result, map outputs are most reliable for broadscale planning and should be interpreted with care at parcel scale unless inputs and choices are made explicit.

This perspective clarifies why buyer due diligence currently feels manual and uncertain: listing sites surface market metadata, while suitability assessments require landscape-aware, indicator-driven spatial analysis. In practice, a prospective buyer must stitch together multiple external sources—soil and climate layers, water access records, and zoning/constraint overlays—to approximate an evidence base. The literature confirms the need for such multi-source synthesis. Spataru et al., working in peri-urban Melbourne, show that agricultural planning rests on assembling heterogeneous datasets and engaging stakeholders to make sense of competing land-use pressures; their findings highlight how agricultural potential and market forces collide in data-sparse, rapidly changing interfaces between urban and rural systems. Although their case targets metropolitan edges rather than farm-dominant regions, it underscores the institutional and informational fragmentation that buyers face when trying to align investment with biophysical opportunity.

Tasmanian planning materials point to a complementary governance reality. The Agricultural Land Mapping Project (ALMP) and the associated Decision Tree and Zoning Guidelines (Tempest & Keteelaar) were created to help councils delineate Agriculture and Rural Zones. Critically, the guidance acknowledges that the ALMP's generic decision rules and desktop GIS can produce anomalies, and it recommends external expert review where automated outputs are ambiguous. Zoning decisions are expected to consider land capability, constraints mapping, irrigation resources, and connectivity, with explicit references to statewide GIS layers (e.g., "Land Potentially Suitable for Agriculture") and to water-resource portals (e.g., WIST, groundwater bores) for checking irrigation feasibility. These materials emphasise that sound land-use allocation depends on integrating formal spatial datasets and policy criteria—precisely the kinds of layers and logic that listing platforms do not provide natively.

Taken together, the literature and planning guidance converge on the same problem your audit exposed: platform-level information is decoupled from biophysical evaluation. Snelder et al. caution that suitability mapping choices (e.g., indicator weights, environmental objectives) introduce subjectivity and that outputs are sensitive to scale and uncertainty; they recommend interactive GIS interfaces so users can inspect indicators and recombine assumptions transparently. In the current ecosystem, listing sites offer neither the datasets nor the analytic controls to satisfy that recommendation, forcing buyers to reconstruct the evidence base outside the marketplace. This gap explains why agribusiness investors experience a manual, error-prone cross-referencing process when attempting to align crop preferences with on-market parcels, and why opportunities can be missed when markets move faster

than due-diligence workflows.

These findings directly motivate the present project's design choice: to embed verified geospatial suitability (soil–climate–topography and irrigation-relevant cues) alongside live property listings, within an interactive, auditable interface. By coupling TESM-derived suitability summaries at the parcel level with links to source layers and planning-relevant resources (e.g., irrigation and constraint overlays), the tool aligns with the literature's call for transparent indicator composition and council guidance on multi-layer evidence. In short, the project aims to transform listing discovery from a seller-narrative space into a data-substantiated decision space, reducing fragmentation and bringing agronomic credibility to property search in Tasmania. Hence, the marketplace exposes availability but not verifiable biophysical suitability, obliging manual cross-referencing.

### 4.6 The Integration Gap

This structural disconnect (modelled suitability on one side, market discovery on the other) defines the integration gap. A consistent theme across the preceding subsections is the structural disconnect between spatial analytics and market-facing property information. On the one hand, land-suitability science (e.g., TESM/DSA) aggregates soil, climate, and terrain into interpretable, rule-based classifications; on the other, PropTech platforms optimise for transaction readiness, valuation signals, and user-generated content. The consequence is that investors, planners, and buyers encounter market data devoid of biophysical evidence, while suitability maps remain institutionally and technically decoupled from live listings. Conceptually, this gap is not merely a UX shortcoming: suitability itself is context-dependent and indicator-composed, not an intrinsic parcel attribute, which means that claims made in listing text without linked indicators, weights, and objectives are inherently fragile. This underscores the need for interfaces that expose indicator logic and uncertainty alongside availability, rather than treating “suitability” as a static label.

Within the suitability tradition, Tasmania's digital soil assessment demonstrates the strengths of transparent rule sets over high-resolution DSM and climate grids, but it also illustrates where uncertainty handling and scenario-sensitivity must be surfaced for decision support. Probabilistic and fuzzy approaches (e.g., simulation of input error, membership functions) provide materially different—and more honest—appraisals of where land sits relative to class thresholds, compared with crisp cut-offs alone; this is particularly salient at parcel scale and in marginal environments where small input errors flip classes. In short, best practice suggests coupling interpretable rules with explicit uncertainty communication to avoid overconfidence when maps are pulled into investment contexts or used in real time. From a governance perspective, data policy and institutional plumbing also impede integration. International guidance emphasises that modern land institutions should leverage digital

interoperability, earth observation, and transparent registries to lower frictions, improve equity, and enable evidence-based planning; yet many registries and platforms still lack the interfaces (or mandates) to publish and consume geospatial suitability layers with provenance and audit trails. Likewise, cadastral consolidation and spatial planning instruments are designed to integrate multi-sector layers (agriculture, water, nature, infrastructure) but are rarely connected to the retail discovery layer where buyers actually act. A durable bridge therefore requires not only technical joining of GIS and PropTech, but also policy alignment around publication standards, indicator disclosure, valuation neutrality, and user rights.

AgriSuit, a software developed jointly by Matrix Geoscience and Monash University (Business Analytics Creative Activity) is a Shiny-based decision-support platform that operationalises the integration of PropTech listing streams with enterprise/crop-suitability mapping. The system maintains a continuously updated backend database that ingests key attributes from online property listings (e.g., address, indicative price range, locality) and links them to cadastral parcels. For each parcel, AgriSuit computes TESSM-derived suitability summaries at the parcel level (including limiting factors) and exposes the indicator logic so that users can trace how soil, climate, and topography contribute to the result. Where feasible, the interface also surfaces uncertainty cues and provides auditable links to source layers, enabling transparent verification.

To reduce noise and improve relevance, the database excludes clearly non-market parcels (e.g., national parks, charitable holdings, government-owned land), thereby focusing attention on properties that are actually available for purchase. The user experience is designed for non-specialists as well as sector insiders: prospective investors can filter by municipality or specific address (when listed) to identify crops that are biophysically compatible with a parcel, while agribusiness users can interrogate limiting factors directly to understand management implications. In effect, AgriSuit broadens the traditional scope of property search beyond residential attributes to include scientifically grounded agronomic potential, and it streamlines due-diligence by consolidating market availability with objective, parcel-level suitability evidence in a single, auditable interface.

## 5 Data Sources

Domain.com.au for property listings (criteria: TAS,  $\geq 10\text{Ha}$ ).

Cadaster (parcel boundaries).

TESM crop suitability rasters (local vs WMS).

Address datasets (for matching).

Optional: subscriber/user preference logic.

Include a table summarising sources (format, size, relevance).

## 6 Methodology

### 6.1 Data Pipeline

Step-by-step: scraping  $\rightarrow$  address matching  $\rightarrow$  parcel overlay  $\rightarrow$  suitability calculation  $\rightarrow$  dashboard.

#### Data Fetch (Domain.com and Supporting Datasets)

The workflow begins with the data fetch stage, where multiple datasets are collected from both online and cadastral sources to build the foundation for analysis. The primary dataset is obtained from [Domain.com.au](https://domain.com.au) using an automated web-scraping process. Each webpage typically displays ten property cards, and the scraper systematically iterates through all pages to extract essential details such as the property ID, full address, price, land area (in hectares), listing type (e.g., sale or lease), and corresponding URL links. Once this raw property data is gathered, it undergoes initial text preprocessing through string manipulation and pattern-matching techniques to extract structured information. For instance, numeric price values are parsed from text strings (e.g., “Price: \$1.2M”  $\rightarrow$  1,200,000), land area values are detected using pattern extraction, and irrelevant characters or HTML tags are removed. These operations ensure the scraped data is uniform and free from noise before further integration.

In addition to the property listing data, several related datasets are obtained from official cadastral and spatial data sources. These include the Municipality datasets, which contain cadastral parcel boundaries and polygon coordinates across Tasmania’s 29 municipalities, and the Address datasets, which provide verified address references linked to property identifiers (PIDs). These spatial and address datasets are crucial for ensuring the accuracy and georeferencing of the property information. Furthermore, the workflow incorporates Environmental Suitability Model (ESM) TIFF maps, which depict the crop suitability or feasibility for various crops across Tasmania at the pixel level. These



raster layers are integral to later stages of the workflow, where they are spatially overlaid with property polygons to determine the crop suitability profile for each property.

It is important to note that the web-scraping methodology has been employed purely for the prototype phase, providing a flexible and rapid means to access publicly available data for proof-of-concept validation. However, for future scalability or commercialisation, it is strongly recommended to transition towards API-based data retrieval, leveraging official endpoints from Domain.com and cadastral data providers. An API interface would ensure higher data accuracy, stability, and compliance while reducing the operational overhead of scraping.

The cleaned and preprocessed property data is then exported to a CSV file and stored within the project workspace as an intermediate staging layer. Finally, the refined data—along with the cadastral and ESM raster datasets—is loaded into the DuckDB database, forming a unified data foundation for subsequent stages such as data modelling, spatial transformation, and suitability analysis.

### **Data Modelling, Cleaning, and Transformation**

After the data acquisition stage, the workflow transitions into the data modelling and transformation phase, where the collected datasets are standardised, validated, spatially structured, and then processed to generate crop suitability metrics for each property. This phase represents the bridge between the raw input data and the analytical outputs that drive the final visualisation and decision-making components.

The process begins with data modelling and cleaning, where the various input datasets — including the web-scraped property listings, cadastral parcel data, and verified address information — are integrated into a unified relational database structure. As shown in the data model (Figure X) and the Entity Relationship (ER) diagram (Figure Y), the schema consists of five key interconnected tables: Municipality\_Master, Address\_Master, Property\_Master, Property\_Listing, and CROP\_PROPERTY\_COUNT\_MASTER. The Municipality\_Master table holds cadastral parcel data for all 29 Tasmanian municipalities, containing fields such as PID, CID, and Municipality\_ID, along with the polygon coordinates that define each property's spatial boundaries. These coordinates form the geospatial foundation for the analytical operations carried out later. The Address\_Master table provides verified address references linked via PID and GEOCODE\_ID, while the Property\_Listing table stores the Domain.com property data, with FULL\_ADDRESS serving as the key joining field. The Property\_Master table acts as a relational hub that consolidates all identifiers and establishes unified records by linking PID, FULL\_ADDRESS, and Municipality\_ID.

Before the relational model is established, all datasets undergo a rigorous data cleaning and validation process. Duplicate entries are removed, null or incomplete fields are resolved, and inconsistent

naming conventions across the municipality and address data are standardised. To ensure consistency across all datasets, the address components — including street name, suburb, and postcode — are concatenated into a single `FULL_ADDRESS` field. This concatenation creates a uniform join key that allows for accurate integration between the `Address_Master` and `Property_Listing` tables. Spatial validation checks are also performed to confirm that the geometries in the `Municipality_Master` table align with the corresponding address and property records. By the conclusion of this stage, a clean, standardised, and fully georeferenced relational database is established, ensuring efficient and accurate joins between all datasets.

Once the property and cadastral data are successfully joined, the workflow progresses to the data transformation stage, where the geospatial analysis is performed to derive crop suitability counts for each property. The polygon geometries from the `Municipality_Master` table are spatially overlaid on the Environmental Suitability Model (ESM) raster maps — georeferenced TIFF layers that depict the crop feasibility across Tasmania. Each raster layer contains pixel-level classifications that indicate varying degrees of suitability for specific crops, such as Highly Suitable, Moderately Suitable, Marginally Suitable, or Unsuitable. Using spatial intersection and aggregation techniques, the system identifies all pixels that fall within each property's polygon boundary and counts the number of pixels associated with each suitability class. This process effectively quantifies how much of each property's total land area falls under different suitability levels for each crop type.

The resulting output from this transformation is a structured dataset that links every Property ID (PID) and Full Address with multiple crop types and their corresponding pixel counts across suitability classes. This processed information is then stored in the `CROP_PROPERTY_COUNT_MASTER` table — the final analytical table in the database schema. This table provides a property-level summary of crop feasibility across Tasmania and serves as the foundation for the RShiny dashboard. Through this integration, users can interactively filter, visualise, and analyse agricultural suitability by crop, municipality, or class label, completing the end-to-end data flow from raw property listings to actionable spatial insights.

**Don't forget to Insert diagram (supervisor specifically asked)**

**Mention database structure (DuckDB or equivalent).**

## 6.2 Suitability Computation

**How these crops are determined suitable**

Tasmania's Enterprise Suitability Maps (TESM) are modelled using a deterministic, rule-based framework grounded in digital soil assessment (DSA). The soil component relies on machine learning (interpolation) and geostatistical methods to predict soil attributes from field observations and envi-

ronmental covariates while quantifying uncertainty. Climate and terrain factors are integrated through threshold rules derived from agronomic literature, expert workshops, and industry consultation.

The process uses a “most-limiting-factor” approach. This means that, even if most factors are excellent, just one poor factor can determine the overall suitability. In other words, the weakest link sets the limit:

**Suitability levels are usually scored or classified from:**

- **1.0 Well suited:** All individual criteria are “Well suited.”
- **2.0 Suitable:** “The lowest category among the criteria is Suitable, with all others rated higher.”
- **3.0 Moderately suitable:** One or more “Moderately suitable” factors.
- **4.0 Unsuitable:** Any factor is “Unsuitable” (or the land is protected, urban, or open water).

If a location scores “Well suited” for everything except drainage (which is “Unsuitable”), the whole site is deemed “Unsuitable” for that crop, because the poor drainage is expected to severely limit success.

**Improving Suitability** - If a limiting factor is considered manageable (a “manageable constraint”), landowners can take steps to address it. For example:

- **Low soil pH :** Applying lime can raise pH, making the soil suitable for crops like barley.
- **Poor drainage :** Installing drainage systems can help.
- **Excess stones:** Stones can be mechanically or manually removed.

If such mitigations are feasible, the suitability rating may improve from “Moderately suitable” to “Suitable (with management).”

Suitability is determined by rating all important growing conditions, then choosing the lowest rating among these (the most limiting factor) as the overall score. If that limiting factor can be addressed by practical management techniques, the land’s suitability rating for a crop might be increased to reflect that potential improvement.

In summary, TESM combine expert-defined deterministic rules with advanced digital soil mapping to provide a practical, interpretable, and scientifically robust tool for land-use planning, acknowledging inherent limitations in data and modelling assumptions (Kidd et al., 2012; Kidd et al., 2014; Webb et al., 2014).

Majority pixel rule (explain rationale of the  $\geq 50\%$ ).

Alternative approaches (weighted average, thresholds).

### 6.3 Dashboard Development(AgriSuit Application)

Framework: R Shiny.

#### Purpose

AgriSuit operationalises parcel-level suitability by exposing TESM-derived indicators and listing metadata through a minimal, auditable interface designed for non-specialists. The goal is reproducible, decision-ready outputs with clear provenance and bounded query scope.

#### Architecture

AgriSuit follows a three-layer design that separates data preparation, service orchestration, and interactive rendering.

- Data layer. TESM rasters are pre-summarised to parcel-level indicators (majority suitability class; limiting factors). Listing metadata (address, locality, indicative price) is ingested and reconciled to cadastral parcels; non-market parcels (e.g., national parks, charitable holdings, government land) are filtered upstream.
- Service layer. Parameterised queries handle geocoding/matching, apply validation rules, and stamp results with provenance (dataset version, refresh timestamp).
- UI layer (R Shiny). Reactive modules manage filters, a unified results set (table + map), and state (validation messages, row caps, legends).

#### Core features

---

| Feature         | Details                                                                                                                                                                                              |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filtering       | Crop Category; Crop; Class Label; Municipality; Full Address; Price Min; Price Max; Maximum Rows.                                                                                                    |
| Validation      | Inputs must satisfy (Crop OR Class Label) AND (Municipality OR Full Address); ambiguous or under-specified queries are rejected with prompts.                                                        |
| Results (Table) | Parcel ID; majority suitability class; limiting factors; listed price; direct link to the public listing.                                                                                            |
| Results (Maps)  | <ul style="list-style-type: none"><li>• Context map (WMS) situating the parcel within Tasmania.</li><li>• Parcel-detail map with pixel-level suitability and a crop-specific class legend.</li></ul> |
| Registration    | Email (required); Full Name; Phone; at least two preferences from Preferred Crop; Preferred Municipality; Class Label; Price Range (min/max);   Alert Frequency.                                     |

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Inputs (crop preference, filters).

Outputs (listing table, map overlay, “go to website” button).

## 7 Results

### 7.1 Functionality Showcase

Show examples of scraped listings.

Overlay results (parcel + TESS suitability).

Dashboard screenshots.

### 7.2 Case Example(s)

One or two properties as illustrations: “This property in NW Tasmania is 80% suitable for barley, 60% for carrots.”

## 8 Discussion

Strengths: first tool combining TESS with live property data.

Challenges:

Address matching issues.

Local rasters vs WMS (computational trade-off).

Crop coverage (cherries/hazelnuts missing).

Opportunities for improvement:

Add more crops.

Automate subscriber notifications.

Link climate projections.

## 9 Conclusion

Restate contribution: bridging TESS with real-estate.

Highlight impact: empowers faster, data-driven agribusiness investment.

Point to future expansion: broader crops, live updates, policy use.

## 10 Resources

- **TESM:** <https://nre.tas.gov.au/agriculture/investing-in-irrigation/enterprise-suitability-toolkit/enterprise-suitability-mapsy>
- **TESM Modelling References:**
- **NZ S-map:** <https://smap.landcareresearch.co.nz/>
- **FAO EcoCrop:** <https://ecocrop.fao.org/>